

Prerequisites

Python 3.8+

pip (package manager)

Git installed

Gemini API Key (free)

01

Clone the Repository

Download the DebtLens source code to your local machine using Git.

```
git clone https://github.com/passionate-coder26/DebtLens---Anti-Pattern-Detector-Technical-Debt-Intelligence.git
cd DebtLens---Anti-Pattern-Detector-Technical-Debt-Intelligence/backend_engine
```

02

Install Dependencies

Install all required Python packages listed in requirements.txt.

```
pip install -r requirements.txt
```

■ Packages installed: flask, flask-cors, gitpython, radon, joblib, pandas, numpy, scikit-learn, google-generativeai

03

Set the Gemini API Key

Export your Gemini API key as an environment variable. Get a free key at aistudio.google.com.

```
# Windows
set GEMINI_API_KEY=your_gemini_api_key_here

# Mac / Linux
export GEMINI_API_KEY=your_gemini_api_key_here
```

04

Start the Flask Server

Launch the backend API server. It will serve both the REST API and the frontend UI.

```
python api.py
```

■ Server starts on <http://127.0.0.1:5000> by default.

05

Open the Application

Navigate to the URL below in your browser to access the DebtLens dashboard.

```
http://127.0.0.1:5000
```